Inverse Problems in Geophysics
Part 12: Global minimization

2. MGPY+MGIN

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Evaluation

- only 6 participants (5GP + 1GI + 1PhD)
- in summary rather good, but points to be improved
- hard conditions: hybrid & english course, different levels
- particularly exercises: separate online/inplace? additional tutor?

Content

Question	Totally	Largely	partly	largeDis	disagree
significant process	2	2	1	1	0
increase interest	2	3	0	0	1
clear objectives	3	2	1	0	0
well structured	4	1	1	0	0
lack of knowledge	1	1	1	1	2

- Which knowledge is required? Python? Math? EM?
- How to improve clarity, progress and interest?

Level

Topic	too low	perfect	too high
requirements	0	5	1
quantity	0	6	0
tempo	0	4	2

- tempo: I know. More summaries?
- content: I agree (for lectures), exercises focus on less points
- progress in exercises: very different
- clearer exercise structure: 10min intro + free programming
- not sure if TEM was a good idea (before: VES or MT)

Lecturer

Question	Totally	Largely	partly	largeDis	disagree
availability	6	0	0	0	0
questions	6	0	0	0	0
complex	2	1	1	1	1
motivating	5	0	0	1	0
understandable	3	2	0	1	0
language skills	5	1	0	0	0
material helpful	4	1	1	0	0
slides&notes	4	0	1	1	0

Special points

- distracted: I realized,
- less use of the interactive board (online audience?)
- exercises: chaotic and contradictionary
- code parts: less explanations,

What I learned:

- better preparation of exercises
- slower progress (and less content), get more direct feedback

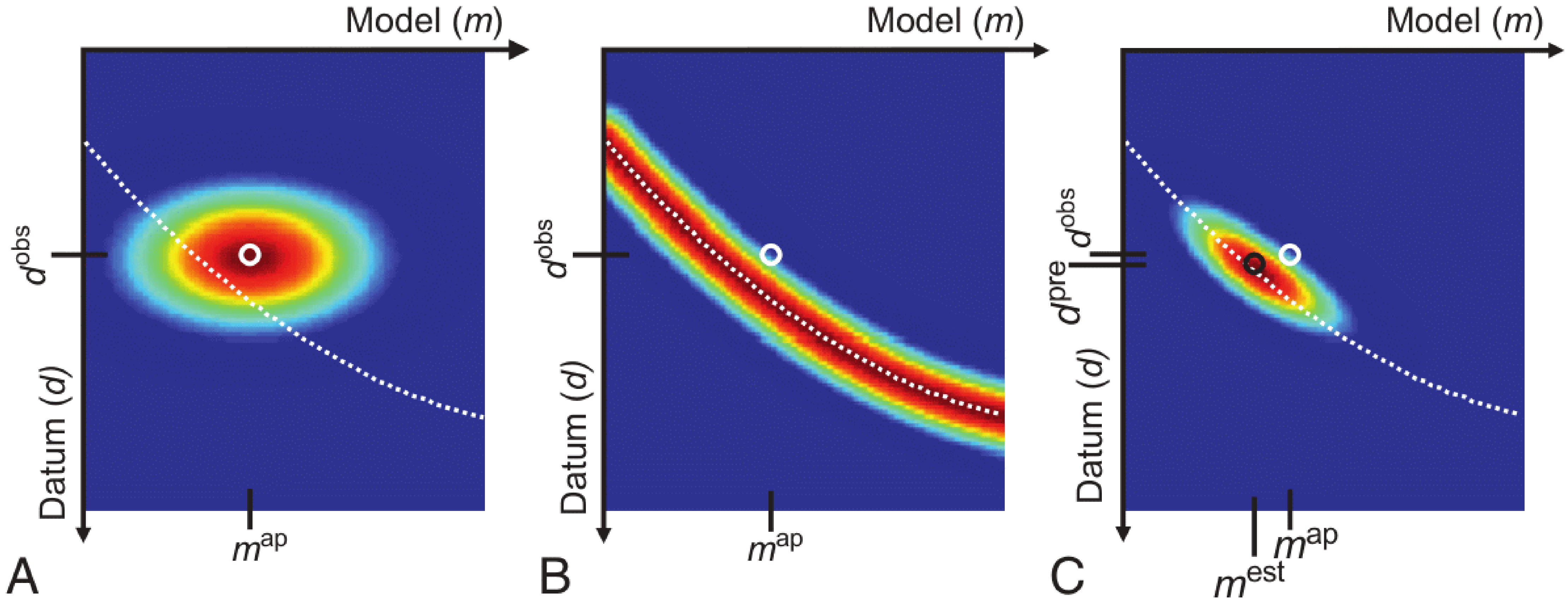
Recap Bayes Theorem

Bayes theorem

$$p(\mathbf{m}|\mathbf{d}) = \frac{p(\mathbf{d}|\mathbf{m})p(\mathbf{m})}{p(\mathbf{d})}$$

- $p(\mathbf{m}|\mathbf{d})$ - posterior probability that model $\mathbf{m}$ is responsible for data
- $p(\mathbf{d}|\mathbf{m})$ - (theoretically) likelihood that $\mathbf{m}$ is capable to generate d
- $p(\mathbf{m})$ - probability that model $\mathbf{m}$ is the actual model
- $p(\mathbf{d})$ - probability of observing $\mathbf{d}$ amongst all possible

Data-model dependence

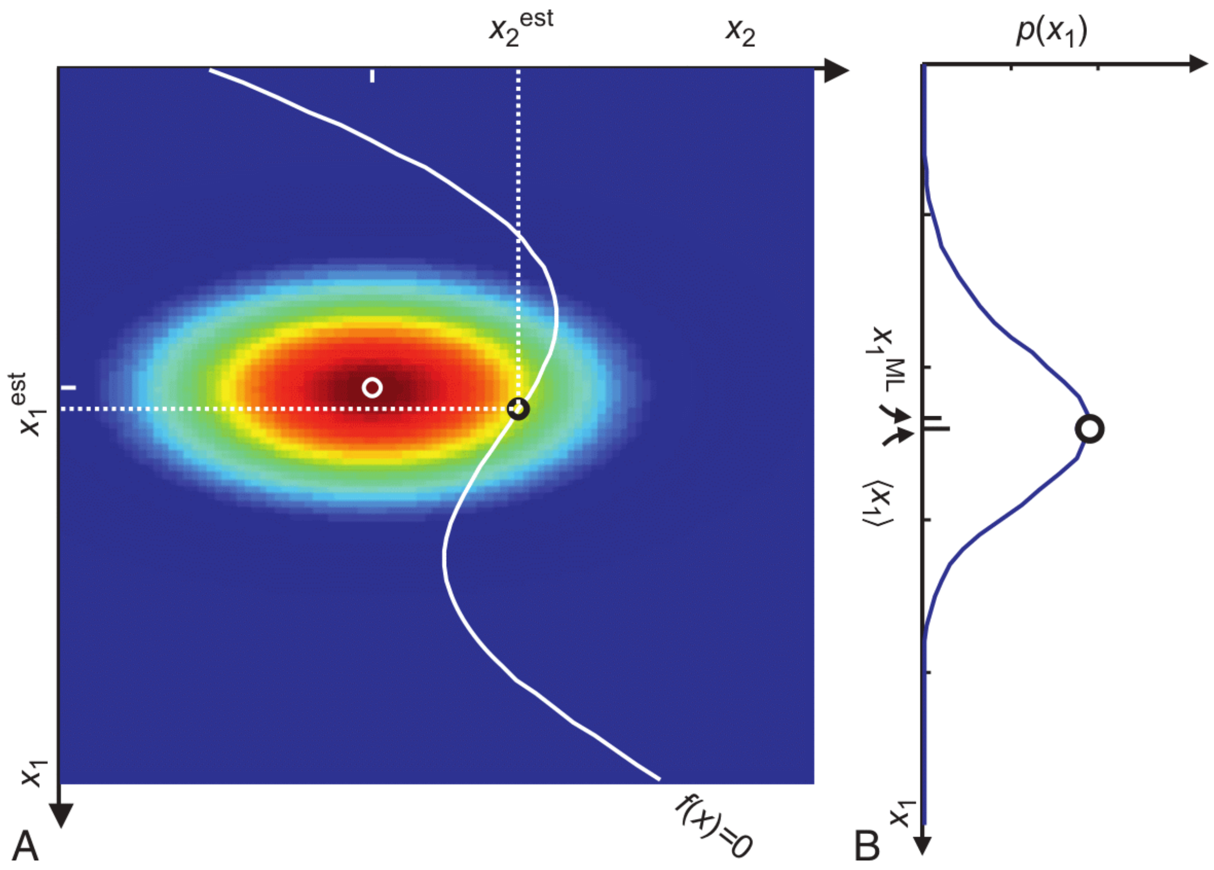


A: *a priori pdf* $p_a(\mathbf{m}, \mathbf{d})$, B: *conditional pdf* $p_g(\mathbf{m}, \mathbf{d})$, C: product $p_t(\mathbf{m}, \mathbf{d}) = p_a(\mathbf{m}, \mathbf{d})p_g(\mathbf{m}, \mathbf{d})$, white: theory

Model space

Bayes theorem provides another view on geophysical inversion

- trade-off between data fit (white curve) and model assumptions (color)
- represented by data errors and regularization terms/strength



Application to constrained inversion

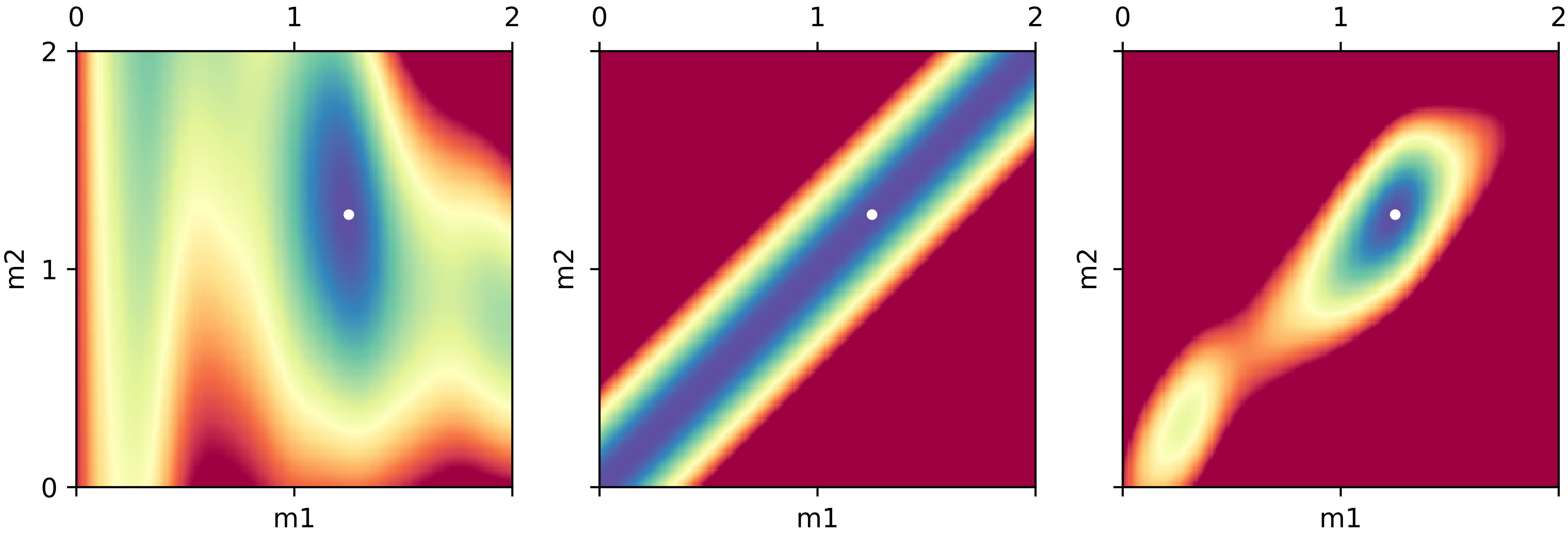
Gaussian likelihood function (data misfit)

$$p(\mathbf{d}|\mathbf{m}) = \frac{1}{(2\pi)^P|D^{-1}|}\exp\left(-\frac{1}{2}\delta\mathbf{d}\mathbf{D}^T\mathbf{D}\delta\mathbf{d}\right)$$

Prior probability function (smoothness of the model)

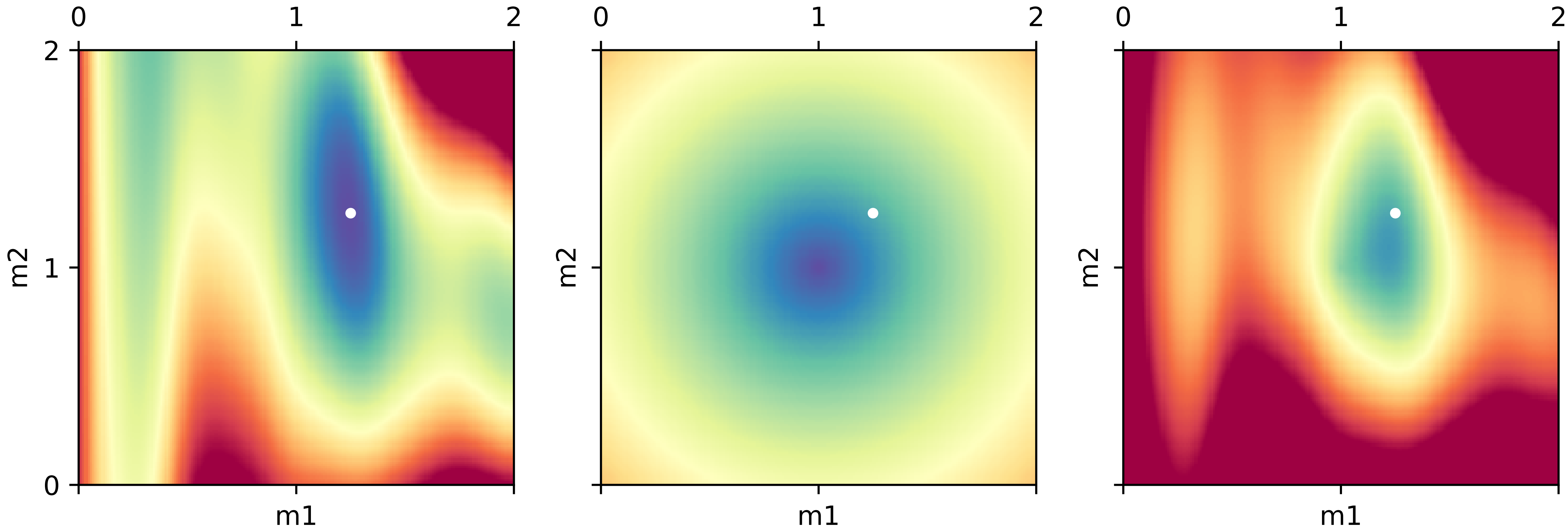
$$p(\mathbf{m}) = \left(\frac{\lambda}{2\pi}\right)^{(N-1)/2}\exp\left(-\frac{\lambda}{2}\mathbf{m}^T\mathbf{C}^T\mathbf{C}\mathbf{m}\right)$$

Smoothness



left: Φ_d, center: $\lambda\Phi_m$, right: $\Phi_d + \lambda\Phi_m$ (λ=300)

Slightly wrong a-priori information



left: Φ_d, center: $\lambda\Phi_m$, right: $\Phi_d + \lambda\Phi_m$ (λ=30)

Global inversion algorithms

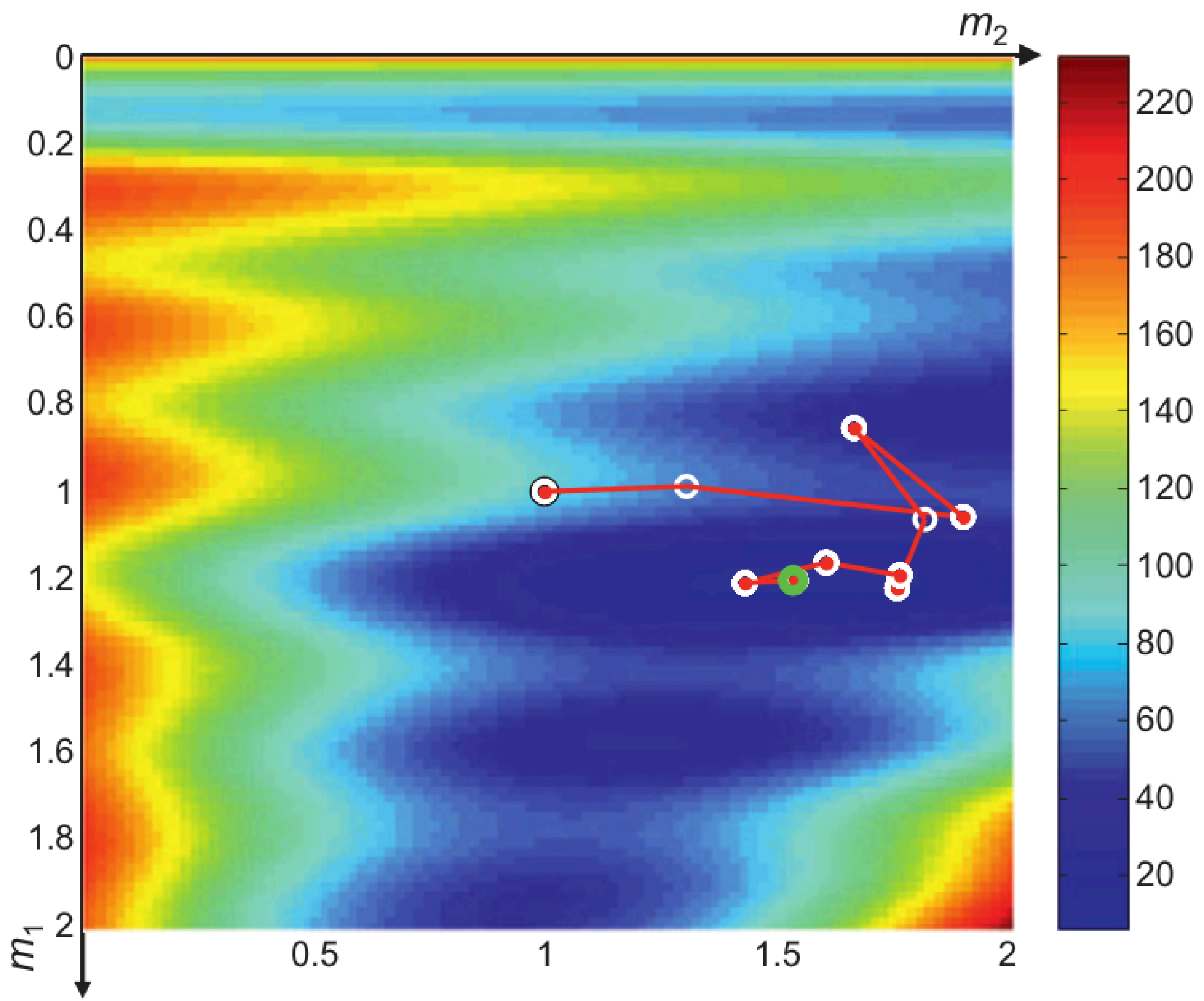
- probabilistic background and with randomness
- good for finding global minima of objective function
- many forward runs needed (only applicable for fast $f(m)$)
- typical for small parameter sizes like curve fitting, 1D inversion

Monte Carlo methods

- Monte Carlos search: randomly draw solutions from grid
- accept solution if better than old
- Markow chain

$$\{\mathbf{m}^0, \mathbf{m}^1, \ldots\}$$

  - $\mathbf{m}^{k+1}$ depends only on $\mathbf{m}^k$
- Metropolis-Hastings (Metropolis et al., 1953; Hastings, 1970)



Monte Carlo method

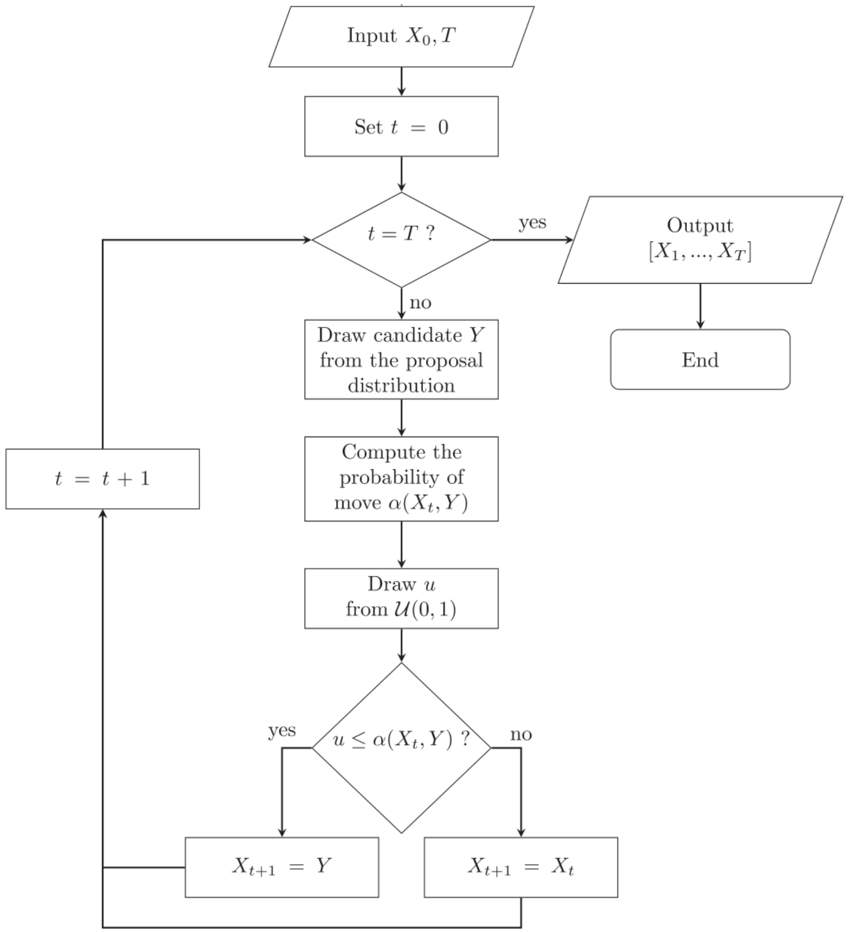
Metropolis-Hastings algorithm

Markov chain

$P(X_{t+1}|X_1,\ldots,X_t) = P(X_{t+1}|P(X_t))$

draw candidate Y from proposal distribution $q(X_t, Y)$ with probability

$$\alpha(X_t, Y) = \min\left[\frac{p(Y)q(Y, X_t)}{p(X_t)q(X_t, Y)}, 1\right]$$



Metropolis-Hastings algorithm

1. Burn-in phase: random walker is methodically guided toward the region of most probable method
2. Drawing of a candidate $\mathbf{m}'$ from a pdf $q(\mathbf{m}|\mathbf{m}')$ that approximates the posterior pdf (e.g., a Gaussian around the current state $\mathbf{m}$)
3. add candidate $\mathbf{m}'$ to Markov chain with acceptance probability

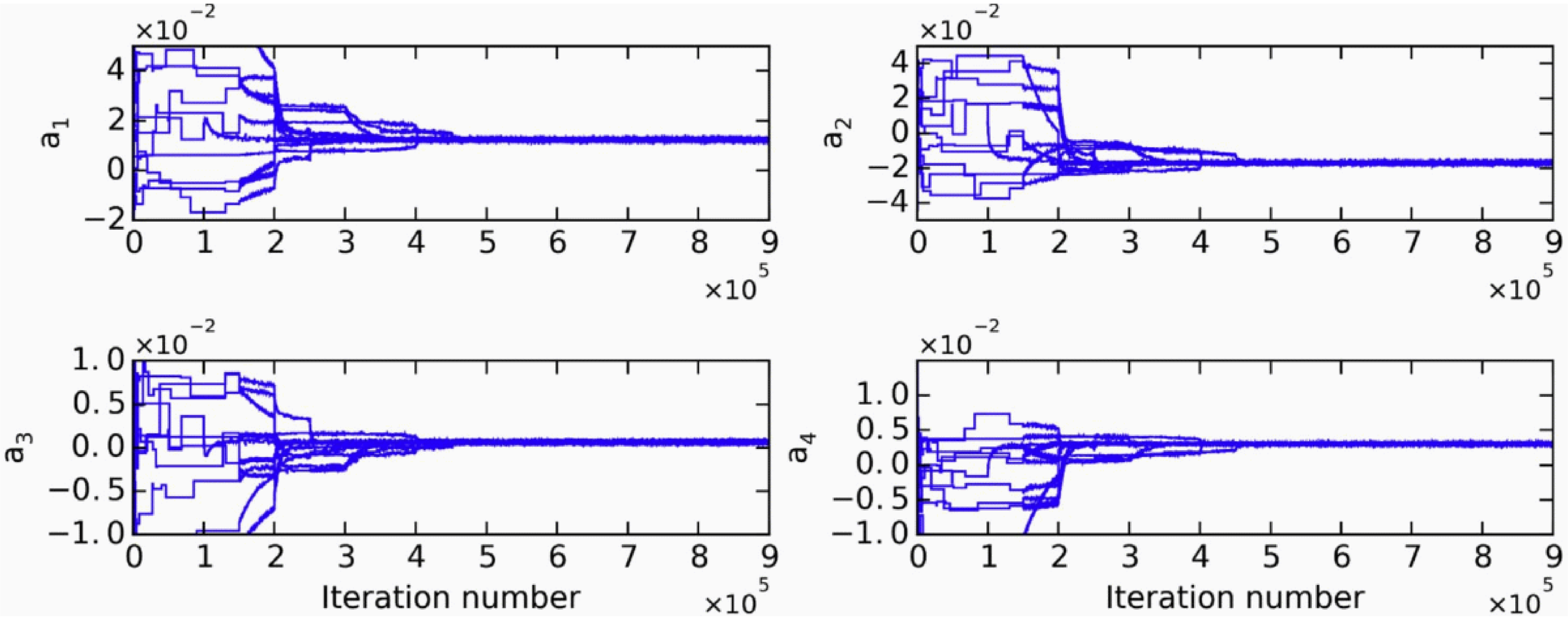
$$\alpha = \min(1, w) \quad \text{with} \quad w = \frac{p(\mathbf{m}') * q(\mathbf{m}|\mathbf{m}')}{p(\mathbf{m}) * q(\mathbf{m}'|\mathbf{m})}$$

using a random variable u in the interval [0, 1] $\Rightarrow$ accept if $u > w$

Key components

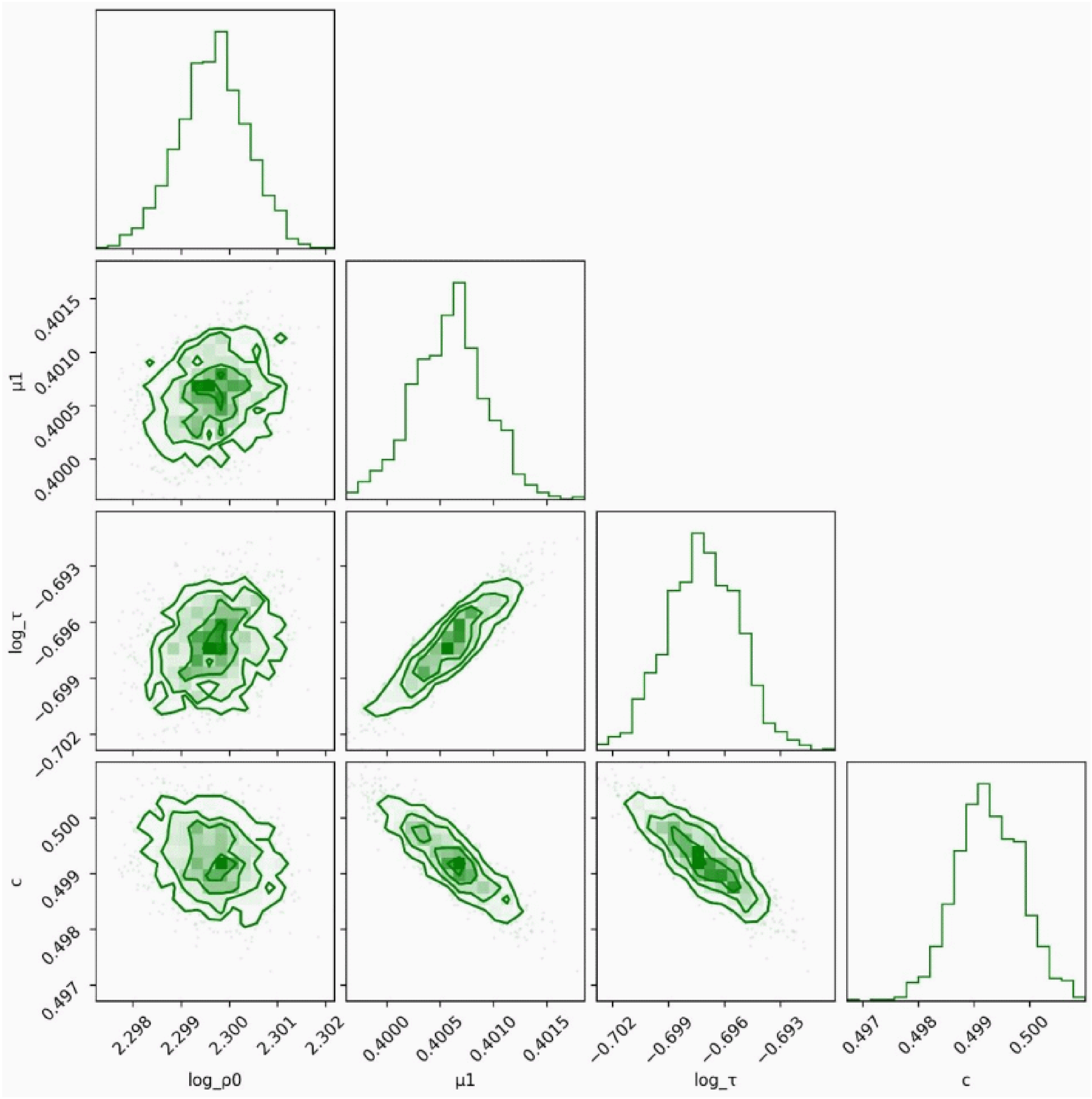
- **Proposal Distribution** (q): Crucial for efficiency. A good proposal distribution explores the space well without being rejected too often.
- **Acceptance Rate**: Good value (20-50%) indicates balance between exploration & exploitation. Too low = slow. Too high = stuck locally.
- **Burn-in Period**: The initial samples often discarded because the chain hasn’t converged to the stationary distribution yet.
- **Thinning**: To reduce autocorrelation between samples, you can keep only every nth sample.
- **Convergence Diagnostics**: Check if chain has actually converged to the target distribution. (e.g., Gelman-Rubin statistic, trace plots).

Trace plots



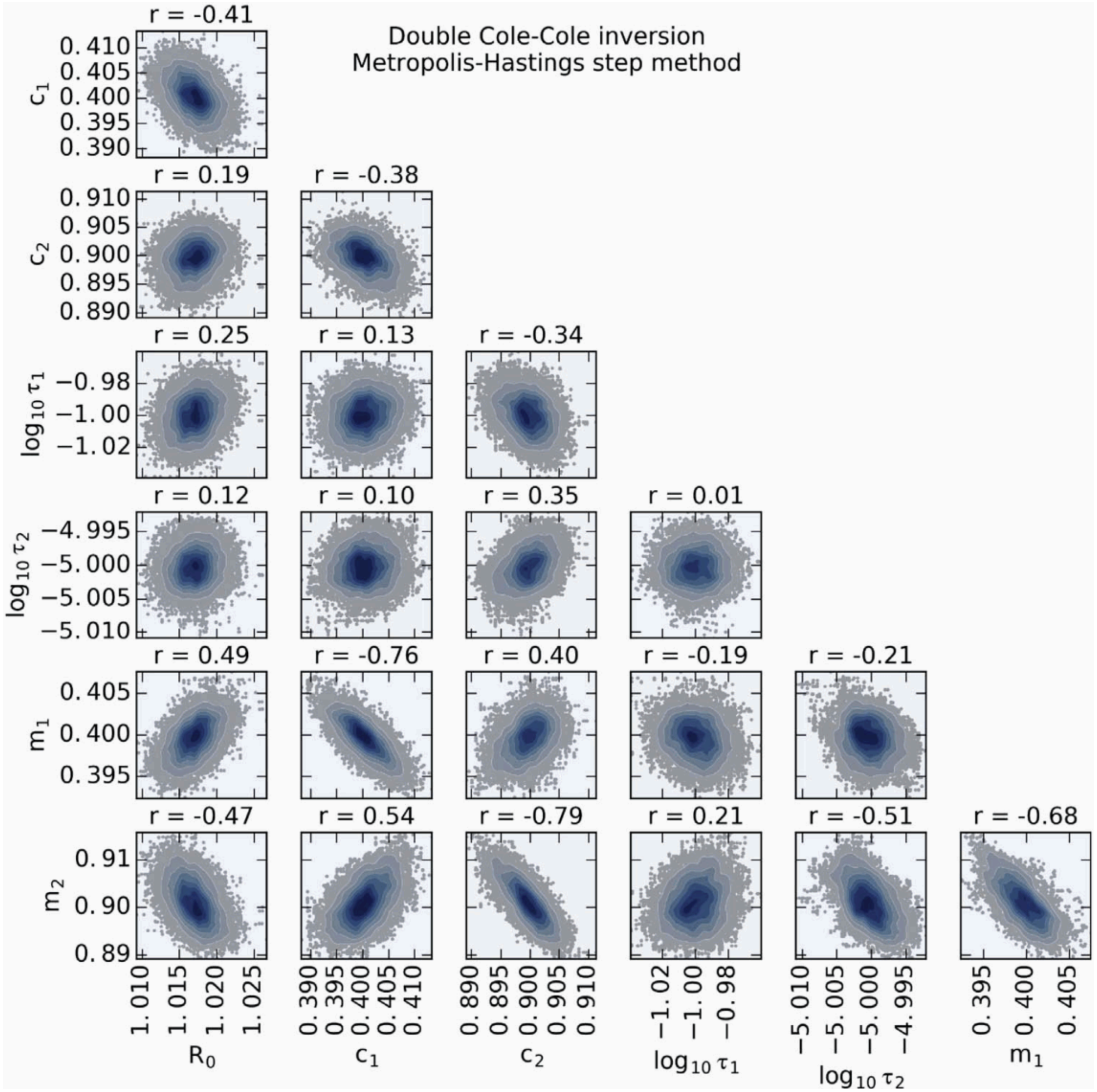
Berube et al. (2017)

Corner plots

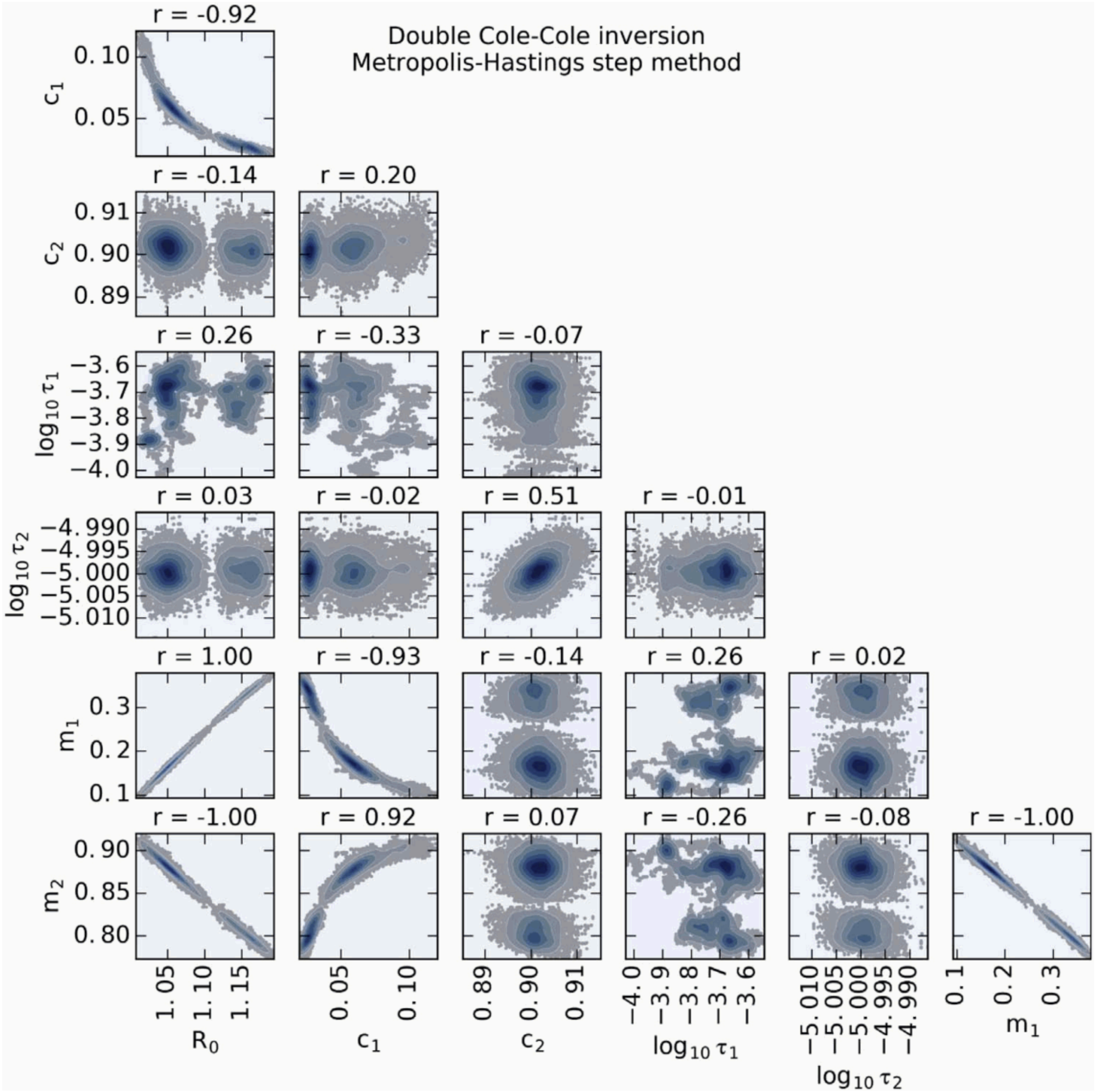


(Roudsari et al., in rev.)

- statistics of models
- correlation of parameters
- uncertainty of the results

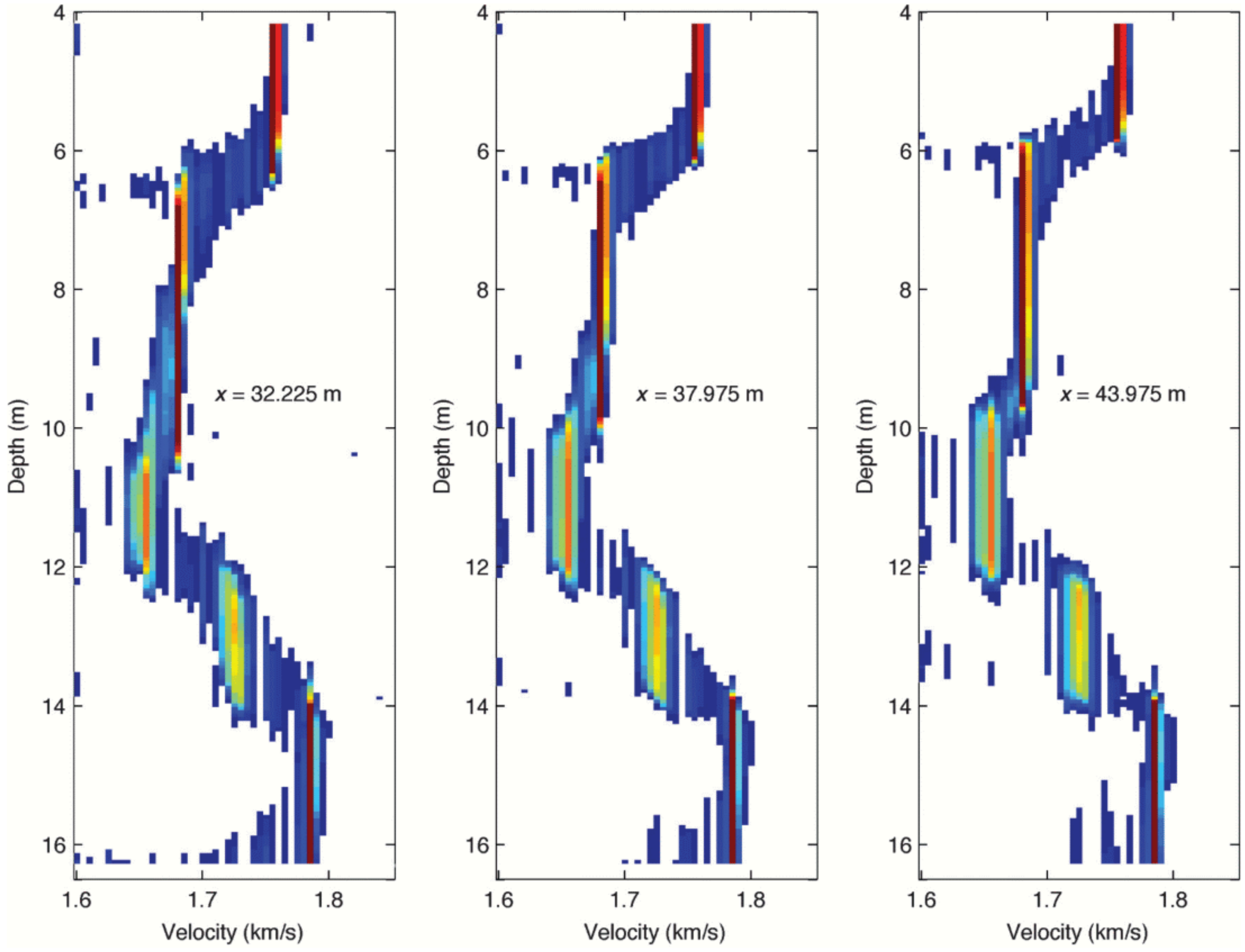


well-defined (Berube et al. 2017)



poorly defined (Berube et al. 2017)

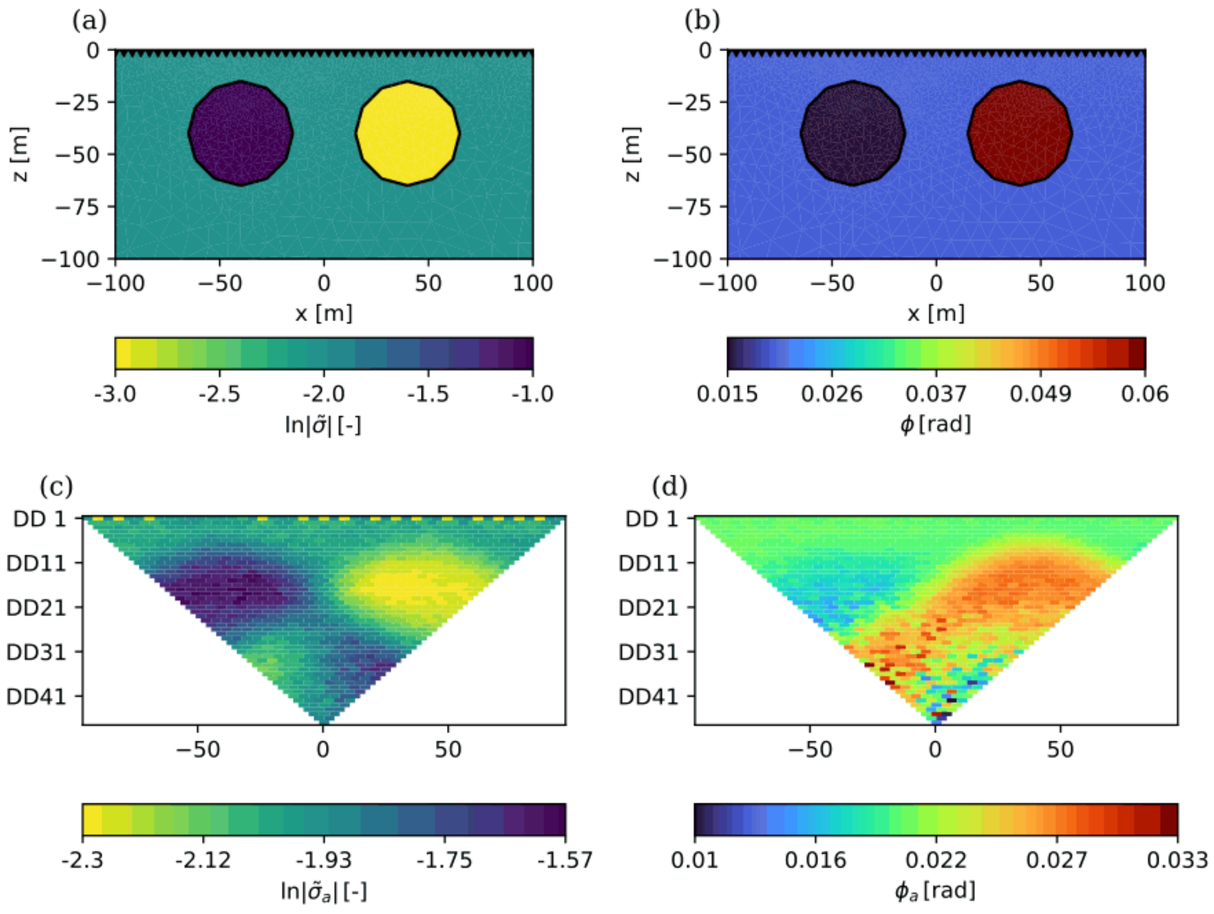
Model variance



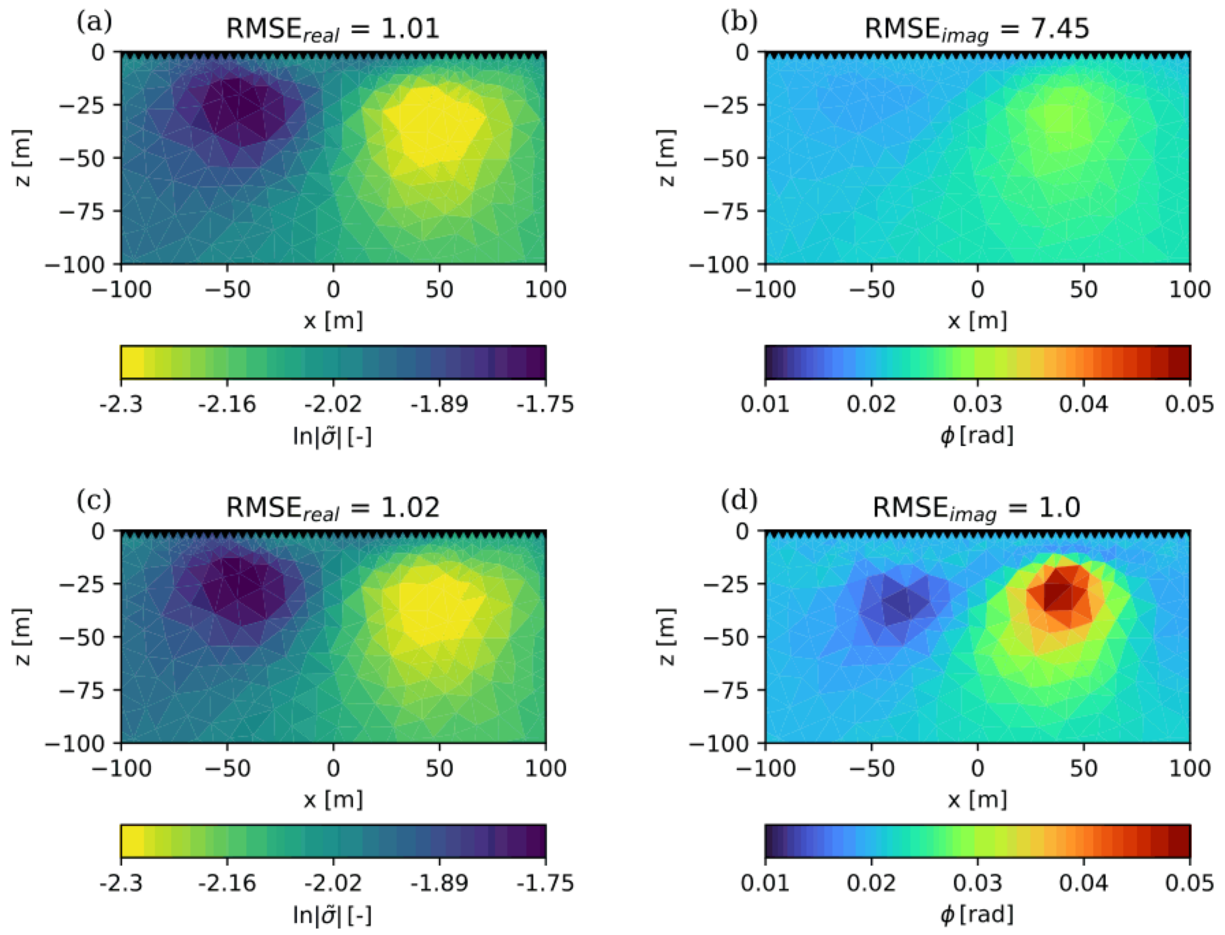
- derivation of mean values and standard deviations
- statistic simulations

(Tronicke et al., 2012)

Application to 2D ERT/IP inversion



Synthetic model & data (Hase et al. 2024)



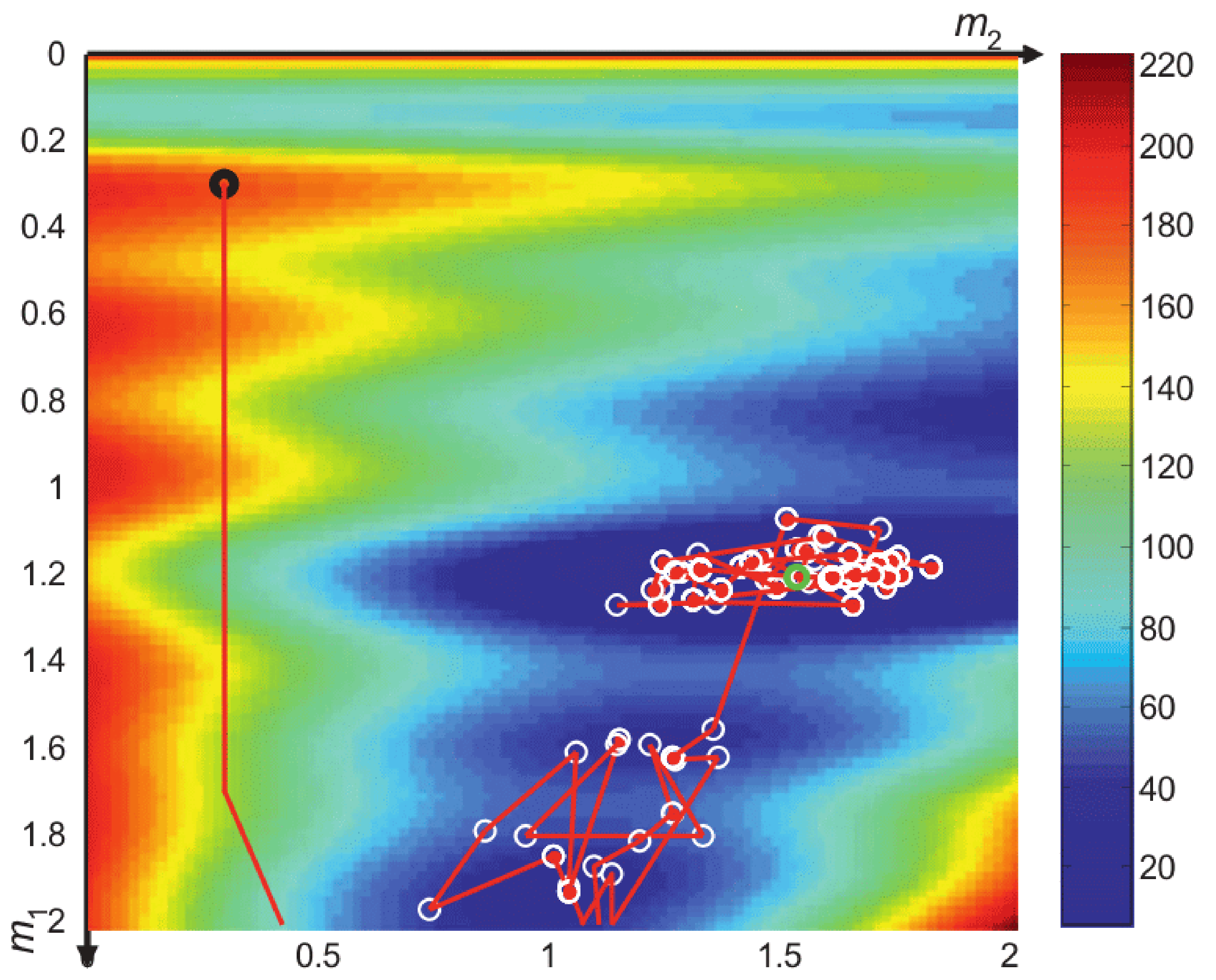
Deterministic (a,b) & probabilistic (c,d) result

Simulated Annealing

Gibbs (Boltzmann) sampling

$$P(E) = e^{-\frac{E}{k_B T}} = e^{-(\Phi(\mathbf{m}) - \Phi(\mathbf{m}^p))/T}$$

- draw models neighboring best fit
- large movements if far from target fit
- small if approaching target



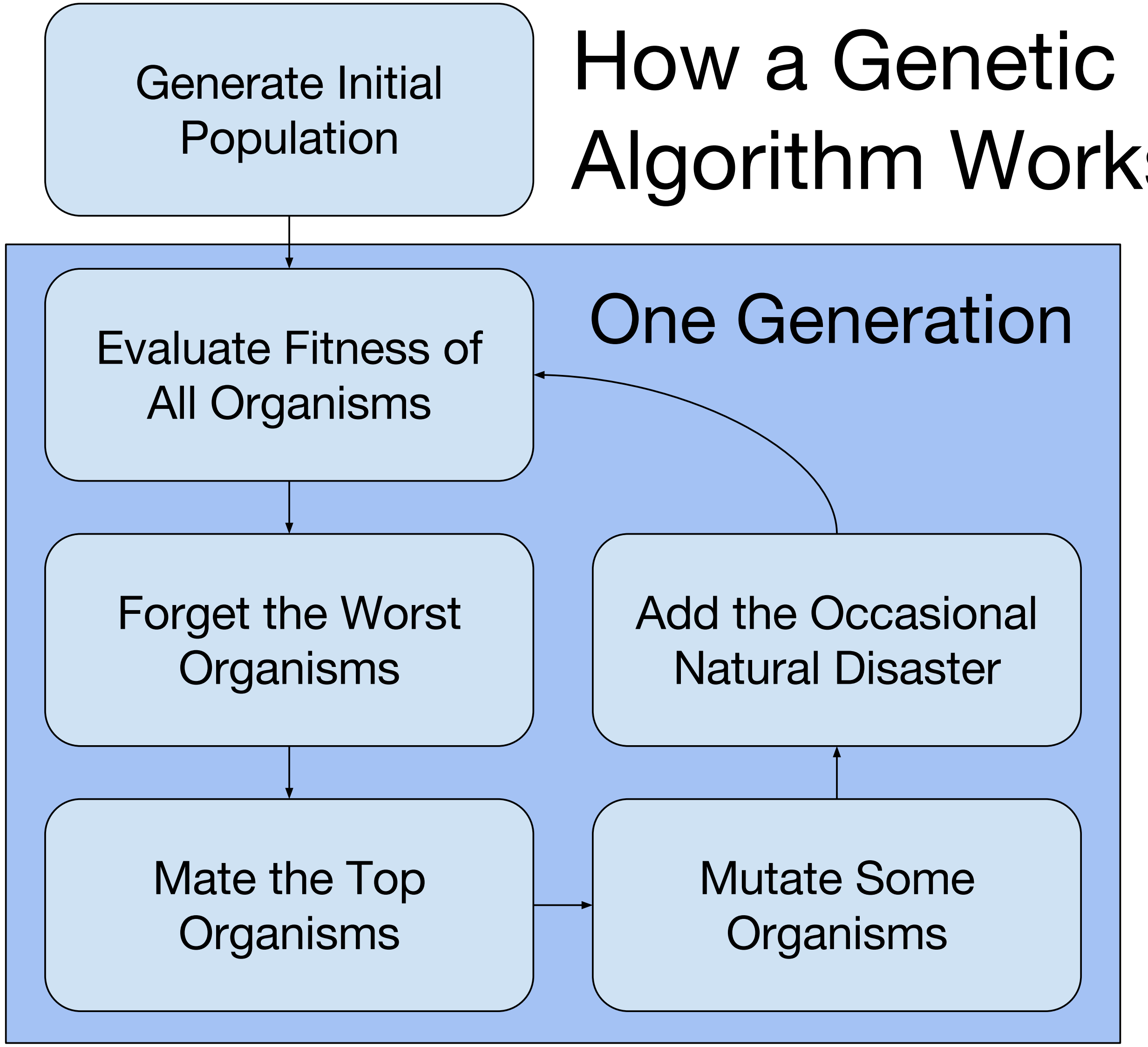
Simulated Annealing

Bio-inspired global search methods

- Genetic Algorithm
- Evolution Strategy
- Differential Evolution Algorithm
- Estimation of Distribution Algorithm
- Particle Swarm Optimization
- Ant Colony Optimization
- Pareto Archived Evolution Strategy (PAES)
- Nondominated Sorting Genetic Algorithm (NSGA-II)

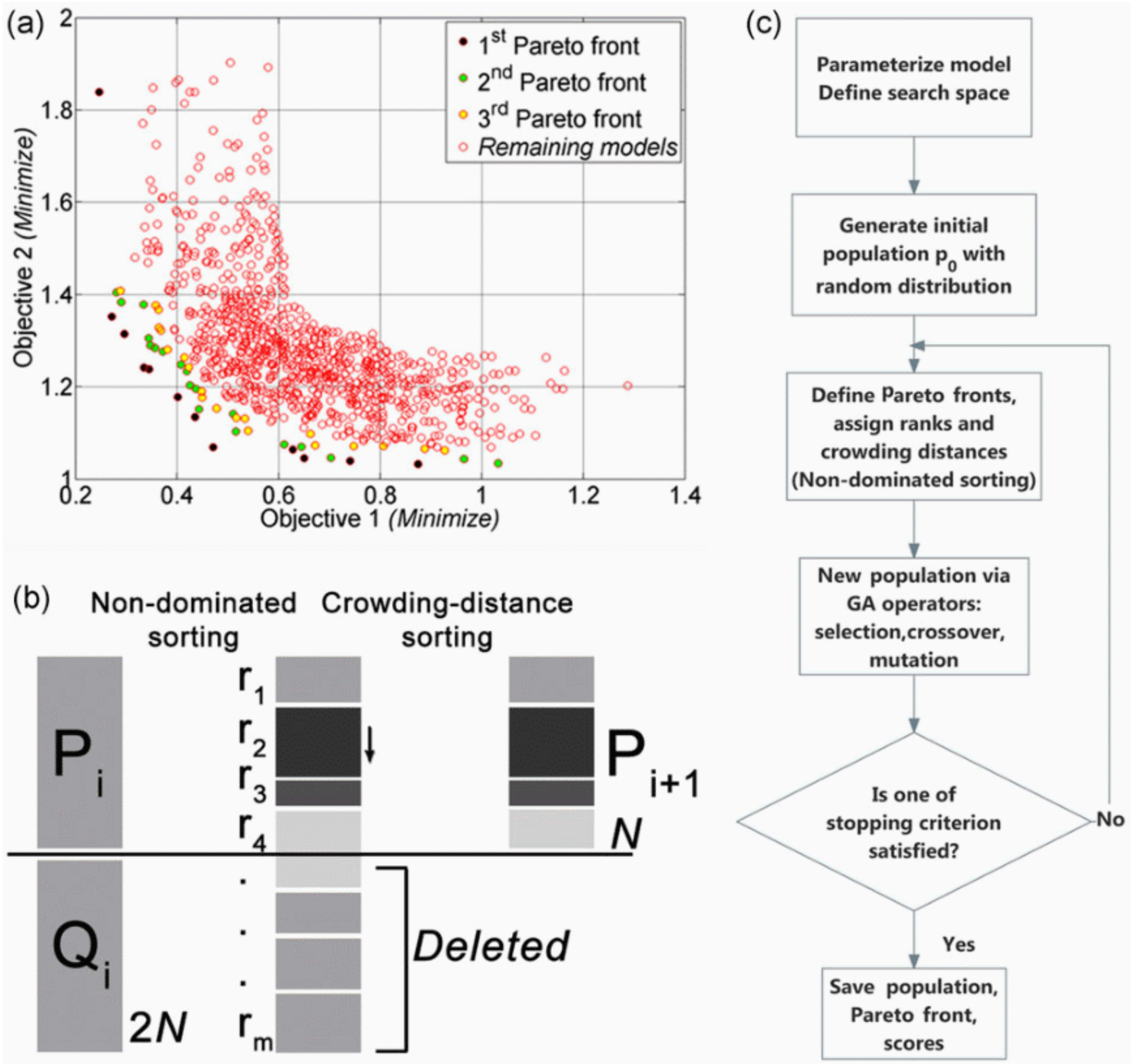
Genetic algorithms

How a Genetic Algorithm Works



- inspired by evolution of species (*survival of the fittest*)
- code models in a binary (DNA) sequence
- randomly generate initial population
- let the fittest (data fit!) survive & mate and the unfittest die
- mutation of the DNA to create variability

Nondominated Sorting Genetic Algorithm

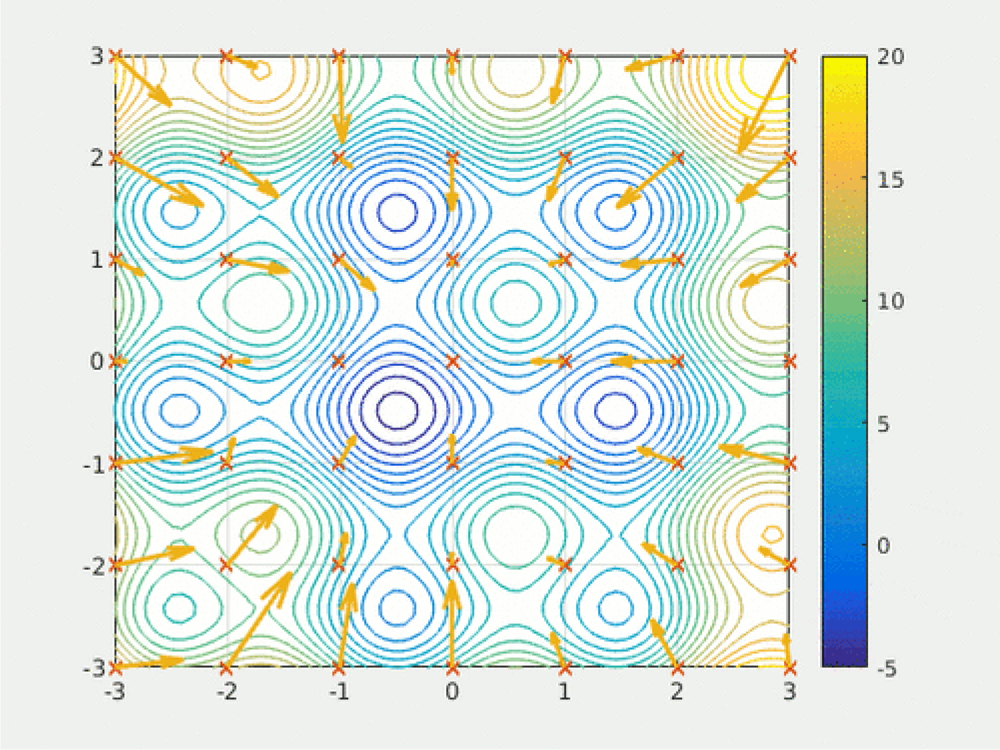


- minimization of two objective functions
  - two geophysical methods
  - data misfit and model roughness
- Pareto front: all non-dominated individuals
- sort Pareto rank and play god

Particle swarm optimization

- movements of individuals in swarm (impulse & attraction)
- individual position ($\mathbf{m}$) & velocity

- distribute population over range
- acceleration from attraction
- move with velocity



Summary

global (probabilistic) inversion methods

- can overcome local minimum problems
- also find a trade-off between data fit and model assumptions
- find a variety of models fitting the data
- can therefore show correlation and model uncertainty
- only applicable to “small” problems (increasingly simple 2D)
- well suited as supplement to local methods